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IT Project Management Support

Lecture 3

Total Cost of Ownership (TCO)

Total Cost of Ownership (TCO)

Based on:

<https://www.cio.com/article/3005705/calculating-the-total-cost-of-ownership-for-enterprise-software.html>

<https://www.business-case-analysis.com/total-cost-of-ownership.html#TCO-covers-ownership-life>

<https://www.techopedia.com/definition/159/total-cost-of-ownership-tco>

<https://www.group50.com/digital-supply-chain-technology-consulting-practice/information-technology-total-cost-ownership/>

<https://digitalprinciples.org/resource/howto-calculate-total-cost-enterprise-software/>

<http://www.softwareadvice.com/tco>

<https://confluence.dimagi.com/display/commcarepublic/Budgeting+for+a+Project>

Total Cost of Ownership (TCO) – definition by Techopedia (1)

<https://www.techopedia.com/definition/159/total-cost-of-ownership-tco>

- Total cost of ownership (TCO) has two connotations, a general definition and a definition that applies to information technology (IT).
- Generally, this definition refers to a financial estimate of all direct and indirect costs associated with a purchased or acquired asset over its entire lifetime, life expectancy or life cycle. It is intended to aid consumers and business entity managers determine the total costs of owning a given product, system or other asset.
- In information technology this refers to a financial estimate of all direct and indirect costs associated with a purchase, capital investment or acquisition of computer hardware and software. Indirect costs include initial installation, personnel training, maintenance, technical support, upgrades and downtime (estimate of business revenue loss).

TCO analysis was popularized by the Gartner Group in 1987. Many different methodologies and software tools have been developed to analyze TCO.

TCO tries to quantify the financial impact of deploying an information technology product over its life cycle. These technologies include software and hardware, and training.

Technology deployment can include the following as part of TCO:

- Computer hardware and programs
- Operation expenses
- Long term expenses.

Computer hardware and programs

- Network hardware and software
- Server hardware and software
- Workstation hardware and software
- Installation and integration of hardware and software
- Purchasing research
- Warranties and licenses
- License tracking - compliance
- Migration expenses
- Risks: susceptibility to vulnerabilities, availability of upgrades, patches and future licensing policies, etc.

Operation expenses

- Infrastructure (floor space lease/rental)
- Electricity (for related equipment, cooling, backup power)
- Testing costs
- Security
- Backup and recovery process
- Technology training
- Downtime, failure, slow processing performance
- Audit (internal and external)
- Insurance
- Information & communication technology personnel
- Executive oversight/management time

Long term expenses

- Upgrades and scalability expenses
- Equipment replacements
- Decommissioning equipment and facilities

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Example: types of whole-of-life costs for a piece of equipment

Purchasing	Operating	Disposal
• market research • specialist advice in assessing business needs and options • developing the business case • the tender process • purchase price of the equipment • other accessories or attachments • delivery costs (freight) and insurance • installation and configuration costs • staff training • initial licenses • warranties • annual insurance • cost of change.	• consumables such as ink cartridges, paper • self- maintenance kits • technical support for users • regular maintenance or servicing • spare parts • energy such as electricity or fuel usage • upgrades • breakdowns and repair costs • cost of replacement service during a breakdown such as hiring another machine or outsourcing the work • IT costs such as hosting • annual insurance • extended warranties • costs of any person who operates or supports the equipment.	• decommissioning costs, possibly involving technical specialists • transportation of the machine away from the work site • fees for disposal of parts, particularly if any are dangerous, for example, cathode vacuum tubes or chemicals • migrating data or removing confidential data • costs of any interim arrangements between different providers • cost of change • site clean-up. Remember to add back any money received on the resale of the equipment!

- **TCO (Total Cost of Ownership)** is defined as all of the possible costs involved in the purchase, installation, management, support and use of the IS infrastructure within an organization throughout its life cycle, from acquisition to disposal.
- Acquisition makes up only 20 - 25% of the TCO on a PC. The rest is made up of tech support, end user support and training, asset management, disposal, playing with technology (**Futz Factor**), and peer support ("**Hey Joe**" support).

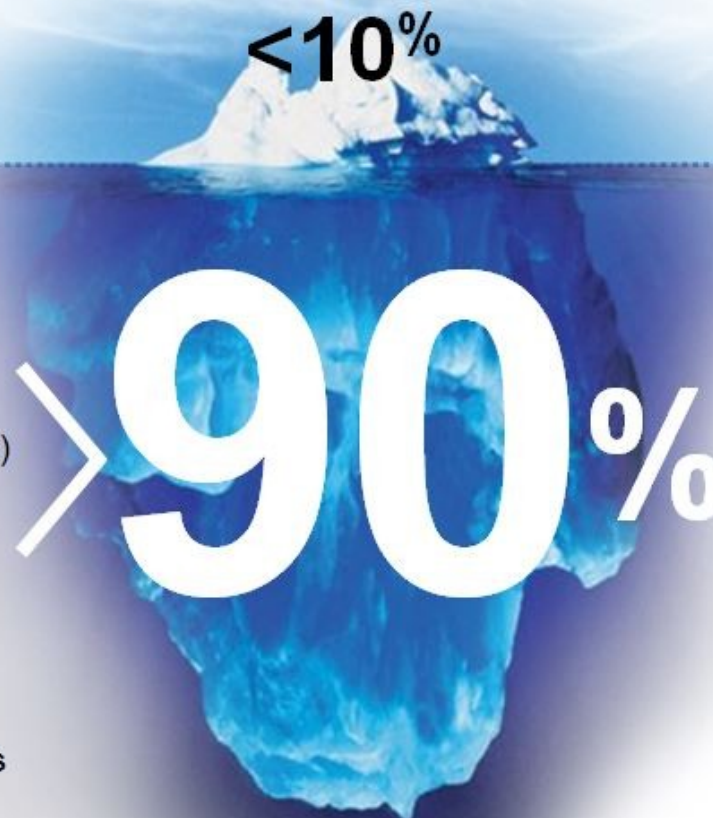
- **The Futz factor:** Wasting time “futzing” (definition: To waste time or effort on frivolities; fool) around with new technology.
- A report from Software Business Technology (SBT) of Sausalito, California, which found that staff spent 5.1 hours each week "futzing" with computers.
- What is the futz factor, you say? It's simple - it is the tendency where people end up wasting a lot of time learning about computers or waiting for their computer to do something.
- It is time spent solving the little, mundane problems that occur day-to-day as they struggle to master their machines.

- **Peer Support or "Hey Joe" support:**
A coworker gives informal assistance to resolve technical problems.
- "Hey, Joe" support is particularly expensive, because if Joe does not know the answer, other coworkers could be drawn into the discussion making three of four people unproductive.

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• **Software license & subscription costs**

- Hardware and networking costs
- Downtime costs (planned and unplanned)
- Upgrades cost
- SLA penalties
- Deployment cost
- Operational support cost (day to day operations)
- Performance costs
- Cost of selection of the vendor software
- Requirements analysis cost
- Developer, admin and end-user training cost
- Application design and development costs
- Cost of integration with other systems
- Quality, user acceptance and other testing costs
- Application enhancements and bug fixes cost
- Replacement costs
- Cost of other risks (including security breaches)



Direct Costs (1)

Direct costs pertain to the **acquisition expenses** or the **cost of buying** the system, and cover all of the following activities:

- **Researching** possible products to buy, which is essentially a labor cost but may also include materials cost, such as purchase of third-party research reports or consultant fees.
- **Designing** the system and all the necessary components to ensure that they work well together. Naturally, this cost component will be higher if a move to a totally different system platform is being considered.
- **Sourcing** the products, which means getting the best possible deal from all possible vendors through solicited bids or market research. It's often sufficient to get a quotation from three vendors (with the cheapest one not necessarily being the best choice).

Direct Costs (2)

- **Purchasing** the product(s), which includes the selling price of the hardware, software, and other materials as negotiated with the chosen suppliers. Include all applicable taxes that might be incurred.
- **Delivering** the system, which includes any shipping or transportation charges that might be incurred to get the product into its final installation location.
- **Installing** the system. Bear in mind that installation also incurs costs in utilities and other environmental—*not* just labor costs. If the installation of the system will result in downtime for an existing system, relevant outage costs must be included. Any lost end-user productivity hours during this activity should also be factored in.

Direct Costs (3)

- **Developing** or customizing the application(s) to be used.
- **Training** users on the new system.
- **Deploying** the system, including transitioning existing business processes and complete integration with other existing computing resources and applications. Include here the costs to promote the use of the new system among end users.

Indirect Costs (1)

Indirect costs address the issues of maintaining availability of the system to end users and keeping the system running, which includes the following:

- **Operations management**, including every aspect of maintaining normal operations, such as activation and shutdown, job control, output management, and backup and recovery.
- **Systems management**, such as problem management, change management, performance management, and other areas.
- **Maintenance** of hardware and software components, including preventive maintenance, corrective maintenance, and general housekeeping.

Indirect Costs (2)

- **Ongoing license fees**, especially for software and applications.
- **Upgrade costs** over time that may be required.
- **User support**, including ongoing training, help desk facilities, and problem-resolution costs. Remember to include any costs to get assistance from third-parties, such as maintenance agreements and other service subscriptions.
- **Environmental factors** affecting the system's external requirements for proper operation, such as air conditioning, power supply, housing, and floor space.
- **Other factors** that don't fall into any of the above categories, depending on the type of system deployed and the prevailing circumstances.



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Total Cost of Ownership (TCO)

Calculation method

<https://www.business-case-analysis.com/total-cost-of-ownership.html>

TCO – Life cycle analysis

- **Total Cost of Ownership (TCO)** is an analysis meant to uncover all the lifetime costs that follow from owning certain kinds of assets.
- Ownership brings purchase costs, of course, but ownership can also bring substantial costs for installing, deploying, operating, upgrading, and maintaining the same assets.
- **For many kinds of acquisitions, TCO analysis finds a very large difference between purchase price and total life cycle costs, especially when viewed across a long ownership period.**

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- Those who purchase or manage computing systems have had a high interest in TCO since the 1980s, when the potentially large difference between IT systems prices and systems costs started drawing the attention of the IT consulting community and IT vendor marketers.
- Competitors of IBM, for instance, used TCO analysis to argue that an IBM computing environment was an overly expensive ownership proposition.
- The five year cost of ownership for major hardware and software systems—from any vendor—can be five to ten times the hardware and software purchase price.

How long is life?

In defining ownership life, owners may consider several "lives" that are in view:

Depreciable life.

- The number of years over which owners charge depreciation expense for an asset is its depreciable life. Accounting standards and local tax laws prescribe what this expense should be each year. Impacts due to this expense include the following:
 - Lower book value for the asset.
 - Lower taxable income (lower profit)
 - Tax savings for owners, due to the lower reported income..

How long is life?

In defining ownership life, owners may consider several "lives" that are in view:

Economic life.

- The number of years for which the asset returns more value to owners than it costs to own, operate, and maintain. When these costs exceed returns, the acquisition is beyond its economic life.

Service life.

- The number of years the asset is actually in service.

Accounting life.

- The number of years the asset impacts the organization's accounts.

How long is life?

- the specified TCO lifespan may also depend on the owner's purpose for the analysis:
 1. When the TCO analysis is performed to support budgeting and planning, or to support strategic decision making, ownership life for the analysis is normally taken as the complete length of **time that ownership has financial impact**:
 - Ownership life begins when the acquisition begins causing costs. This may include costs that occur before the actual arrival or use of the acquisition.
 - Ownership life ends when the asset is no longer causing costs and has no continuing financial impact of any kind. This means that all costs of disposal or decommission have been paid and the asset is no longer carried in an asset account on the company's balance sheet.

How long is life?

2. Alternatively, TCO analysis may be designed to cover an **arbitrarily specified number of years, for example, 3 years, 5 years, 10 years**, and so on. This approach is usual in these situations:

- TCO analysis is performed to choose a vendor from competing proposals, or to prioritize competing Capital acquisition proposals.
- The actual economic life, or service life of the acquisition is uncertain.
- The organization's asset life cycle management policies and practices dictate specific life spans for classes of assets.

Obvious and „hidden” costs

1. Obvious costs in TCO analysis

- Obvious costs in TCO are the costs familiar to everyone involved during planning and vendor selection, such as:
- **Purchase cost:** The actual price paid.
- **Maintenance costs:**
 - warranty costs,
 - maintenance labor,
 - contracted maintenance services
 - or other service contracts.

Obvious and „hidden” costs

2. Hidden costs in TCO analysis

- The so-called hidden costs are the less obvious cost consequences that are easy to overlook or omit from acquisition decisions.
- Costs of this kind can be very large and real, nevertheless.
- All belong in the TCO analysis if they do indeed follow from the decision to own something and they are material (large enough to matter).

„Hidden” costs include (1):

- **Acquisition costs:** the costs of identifying, selecting, ordering, receiving, inventorying, or paying for something.
- **Upgrade / Enhancement / Refurbishing costs.**
- **Reconfiguration costs.**
- **Set up / Deployment costs:** costs of configuring space, transporting, installing, setting up, integrating with other assets, outside services.
- **Operating costs:** for example, human (operator) labor, or energy/fuel costs.

„Hidden” costs include (2):

- **Change management: costs:** for example, costs of user orientation, user training, workflow/process change design and implementation.
- **Infrastructure support costs:** for example, costs brought by the acquisition for heating/cooling, lighting, or IT support.
- **Environmental impact costs:** for example, costs of waste disposal/clean up, or pollution control, or the costs of environmental impact compliance reporting.
- **Insurance costs.**

„Hidden” costs include (3):

- **Security costs:**
- **Physical security**, for example, security additions for a building, including new locks, secure entry doors, closed circuit television, and security guard services.
- **Electronic security**, for example, security software applications or systems, offsite data backup, disaster recovery services, etc.
- **Financing costs**: for example, loan interest and loan origination fees.
- **Disposal / Decommission costs.**
- **Depreciation expense** tax savings (a negative cost).

Model for TCO analysis of an IT system acquisition:

	Acquisition Costs	Operating Costs	Change Costs
Software	Obvious costs	Obvious costs	Hidden costs
Hardware	Obvious costs	Obvious costs	Hidden costs
Personnel	Hidden costs	Hidden costs	Hidden costs
NW & Comm	Hidden costs	Hidden costs	Hidden costs
Facilities	Hidden costs	Hidden costs	Hidden costs

Model for TCO analysis of an IT system acquisition:

- Note that the vertical axis represents IT resource categories, and the horizontal axis represents IT life cycle stages.
- The model design is successful if it achieves two objectives:
- Each axis covers the complete set of categories for that dimension that are useful to decision makers and planners.
- The cost categories will capture the obvious costs but also the less-obvious, or "hidden" costs.

Model for TCOanalysis of an IT system acquisition:

The analysis continues by adding the names of resources to each cell.

Acquisition / Hardware Costs 2nd row, 1st column	Operating Costs / Personnel 3rd row, 2nd column
• Server system purchase • PC system purchase • Engineering workstation purchase • Storage space purchase • Other peripheral hardware purchase	• Administrative labor – Systems operators – Systems programmers – Applications programmers – Network admin labor – Storage management – IT management – Other Admin • Trouble shooting labor • Continuing contract labor • Continuing training (professional)

TCO is based on cash flow cost estimates

Here, for example, are one analyst's estimates for just one cost item (Server System Purchase):

Cash Flow Cost Estimates:		\$ in 1,000s			
Proposed System Acquisition Scenario					
	Year 1	Year 2	Year 3	Total	
Server system purchase	454	84	70	608	
Business as Usual Scenario					
	Year 1	Year 2	Year 3	Total	
Server system purchase	154	64	64	282	
Incremental cash flow (Proposal less Business as Usual)					
	Year 1	Year 2	Year 3	Total	
Server system purchase	300	20	6	326	

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TCO is based on cash flow cost estimates

The example shows cash flow statement structure:

COSTS		Yr 1	Yr 2	Yr 3	
Cash outflows		31 Dec	31 Dec	31 Dec	TOTAL
		2014	2015	2016	
Software					
One-time license - Design SW.....	\$	~	~	~	~
One-time license - Database SW.....	\$	~	~	~	~
One-time license - Admin software.....	\$	~	~	~	~
Annual software maintenance.....	\$	~	~	~	~
Total Software	\$	~	~	~	~
Hardware					
Server system purchase.....	\$	~	~	~	~
PC system purchase.....	\$	~	~	~	~
Engineering WS System Purchase.....	\$	~	~	~	~
Storage space purchase.....	\$	~	~	~	~
Other peripheral HW.....	\$	~	~	~	~
HW maintenance.....	\$	~	~	~	~
Total Hardware	\$	~	~	~	~
Personnel Labor					
Initial training - Design engineers.....	\$	~	~	~	~
Initial training - Manufacturing management.....	\$	~	~	~	~
Design engineering labor.....	\$	~	~	~	~
Manufacturing labor.....	\$	~	~	~	~
IT staff labor.....	\$	~	~	~	~
Total Personnel labor	\$	~	~	~	~
NW & Comms Costs					
WAN costs.....	\$	~	~	~	~
Hosting services.....	\$	~	~	~	~
Continuing training - IT support.....	\$	~	~	~	~
Networking SW and HW operations labor.....	\$	~	~	~	~
NW User help desk.....	\$	~	~	~	~
Total Support Personnel	\$	~	~	~	~
Facilities & Other Costs					
Computer room renovation and security.....	\$	~	~	~	~
Offsite backup and disast recovery service.....	\$	~	~	~	~
Process analysis & mgmt consulting.....	\$	~	~	~	~
Total Facilities and Other Costs	\$	~	~	~	~
Total Costs	\$	~	~	~	~

Next the complicated analysis of cash flow



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Thank You very much
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